

CATS⁺

Competishun Advance

Training Sessions

(Lecture - XIII)

Column-Matching

Column-I

A Let $f(x) = 8x^3 + 3x$. If $\lim_{x \rightarrow \infty} \frac{f^{-1}(8x) - f^{-1}(x)}{x^{1/3}} = a$ then find the value of $(8a)$.

B Let $P = \int_0^{\log_e 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^x + 1} dx$ then find the value of $4e^P$.

C Let $P = \lim_{n \rightarrow \infty} ({}^{2n}C_n)^{1/n}$, then find the value of P .

Column-II

P

0

Q

4

R

11

S

$\ln(4)$

T

e^2

A $\rightarrow$ Q

B $\rightarrow$ R

C $\rightarrow$ Q

A

$$4e^4 = 4e^{\ln(\frac{11}{4})} = 4 \cdot \frac{11}{4} = 11$$

C

$$P = \lim_{n \rightarrow \infty} \left(\frac{2^n}{C_n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{2^n}{\frac{(2n)!}{n! n!}} \right)^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{(2n)(2n-1)(2n-2) \dots (n+1)}{n \cdot (n-1)(n-2) \dots 1} \right)^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \exp \left(\frac{1}{n} \ln \left(\frac{2n \cdot 2n-1 \dots (n+1)}{n \cdot (n-1) \dots 1} \right) \right) = \exp \left(\frac{1}{n} \ln \left(\frac{\frac{2n}{n} \cdot \frac{2n-1}{n} \dots \frac{n+1}{n}}{\frac{n}{n} \cdot \frac{n-1}{n} \dots \frac{1}{n}} \right) \right)$$

$$= \exp \left(\frac{1}{n} \left(\sum_{r=n+1}^{2n} \ln \left(\frac{r}{n} \right) - \sum_{r=1}^n \ln \left(\frac{r}{n} \right) \right) \right)$$

$$= \exp \left(\int_1^2 \ln(x) dx - \int_0^1 \ln(x) dx \right)$$

$$K = e^{\ln(K)}$$

$$= \exp(\ln(K))$$

$$\lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$$

$$= \exp \left(\int_1^2 \ln(x) dx - \int_0^1 \ln(x) \cdot dx \right)$$

$$= \exp \left(\left[x \ln(x) - x \right]_1^2 - \left[x \ln(x) - x \right]_{0^+}^1 \right)$$

$$= \exp \left(\left(2 \ln(2) - 2 - (0 - 1) \right) - \left(0 - 1 - \left(\lim_{x \rightarrow 0^+} x \ln(x) - 0 \right) \right) \right)$$

$$= \exp \left(\ln(4) - 2 + 1 + 1 \right) = e^{\ln(4)} = \boxed{4}$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)} = \frac{0}{\infty} = 0$$

$$\begin{aligned}
 & \textcircled{B} \int_0^{\ln(2)} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^x + 1} \cdot dx \\
 &= \int_0^{\ln(2)} \frac{\overbrace{2e^{2x} + e^x - e^{-x}}^{f'}}{\underbrace{e^{2x} + e^x - 1 + e^{-x}}_f} \cdot dx \\
 &= \left[\ln(e^{2x} + e^x - 1 + e^{-x}) \right]_0^{\ln(2)} \\
 &= \ln\left(\frac{4 + 2 - 1 + \frac{1}{2}}{2}\right) = \ln\left(\frac{11}{4}\right)
 \end{aligned}$$

$$\lim_{t \rightarrow \infty} \frac{t}{\sqrt[3]{8t^3 + 3t}} = \lim_{t \rightarrow \infty} \frac{t}{t \left(8 + \frac{3}{t^2}\right)^{1/3}} \quad \textcircled{A} \quad f(x) = 8x^3 + 3x$$

$$= \frac{1}{2}$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(8x)}{x^{1/3}} = 1$$

$$R. Ans = 1 - \frac{1}{2} = \frac{1}{2}$$

$$a = \frac{1}{2}$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(8x) - f^{-1}(x)}{x^{1/3}} = a$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(8x)}{x^{1/3}}$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(x)}{x^{1/3}}$$

$$f^{-1}(x) = t$$

$$x = f(t)$$

$$x \rightarrow \infty \Rightarrow t \rightarrow \infty$$

$$= \lim_{t \rightarrow \infty} \frac{t}{\sqrt[3]{f(t)}}$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(x)}{x^{1/3}} = \frac{1}{2}$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(8x)}{(8x)^{1/3}} = \frac{1}{2}$$

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(8x)}{x^{1/3}}$$

Ques

If $F(x) = \int \frac{(1+x)\{(1-x+x^2)(1+x+x^2)+x^2\}}{1+2x+3x^2+4x^3+3x^4+2x^5+x^6} dx$

and $F(e^2 - 1) = 2$, then find the value of $\sum_{r=1}^{10} F(e^r - 1)$

$$1+x^2+x^4 \\ \equiv (1-x+x^2)(1+x+x^2)$$

$$F(x) = \int \frac{(1+x) \{ (1-x+x^2)(1+x+x^2) + x^2 \}}{1+2x+3x^2+4x^3+3x^4+2x^5+x^6} dx$$

$$= \int \frac{(1+x) \{ (1+x^2+x^4) + x^2 \}}{x^3 \left(\frac{1}{x^3} + \frac{2}{x^2} + \frac{3}{x} + 4 + 3x + 2x^2 + x^3 \right)} dx$$

$$F(x) = \ln(1+x) + C$$

$$x = e^2 - 1 \\ \downarrow f(x) = 2$$

$$2 = \ln(e^2) + C$$

$$2 = 2 + C \Rightarrow C = 0$$

$$F(x) = \ln(1+x)$$

$$\sum_{r=1}^{10} \ln(e^r - 1 + 1) = \sum_{r=1}^{10} \ln(e^r)$$

$$= \sum_{r=1}^{10} r = \frac{10 \times 11}{2} = \boxed{55}$$

Ans

$$= \int \frac{(1+x)(1+2x^2+x^4)}{x^3 \left(\underbrace{x^3 + \frac{1}{x^3}}_{\text{}} + 2 \underbrace{\left(x^2 + \frac{1}{x^2} \right)}_{\text{}} + \underbrace{3 \left(x + \frac{1}{x} \right)}_{\text{}} + 4 \right)} dx$$

$$= \int \frac{(1+x)(1+x^2)^2}{x^3 \left(\underbrace{x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x} \right)}_{\text{}} + 2 \left(x^2 + \frac{1}{x^2} + 2 \right) \right)} dx$$

$$= \int \frac{(1+x)(1+x^2)^2}{x^3 \left(\left(x + \frac{1}{x} \right)^3 + 2 \cdot \left(x + \frac{1}{x} \right)^2 \right)} dx = \int \frac{(1+x) \cdot \cancel{x^3} \cdot \left(\frac{1}{\cancel{x}} + x \right)^2}{\cancel{x^3} \cdot \left(\cancel{x} + \frac{1}{\cancel{x}} \right)^2 \left(\cancel{x} + \frac{1}{\cancel{x}} + 2 \right)} dx$$

$$= \int \frac{1+x}{x^2+1+2x} dx = \int \frac{1+x}{(x+1)^2} dx = \int \frac{dx}{1+x} = \ln(1+x) + C$$

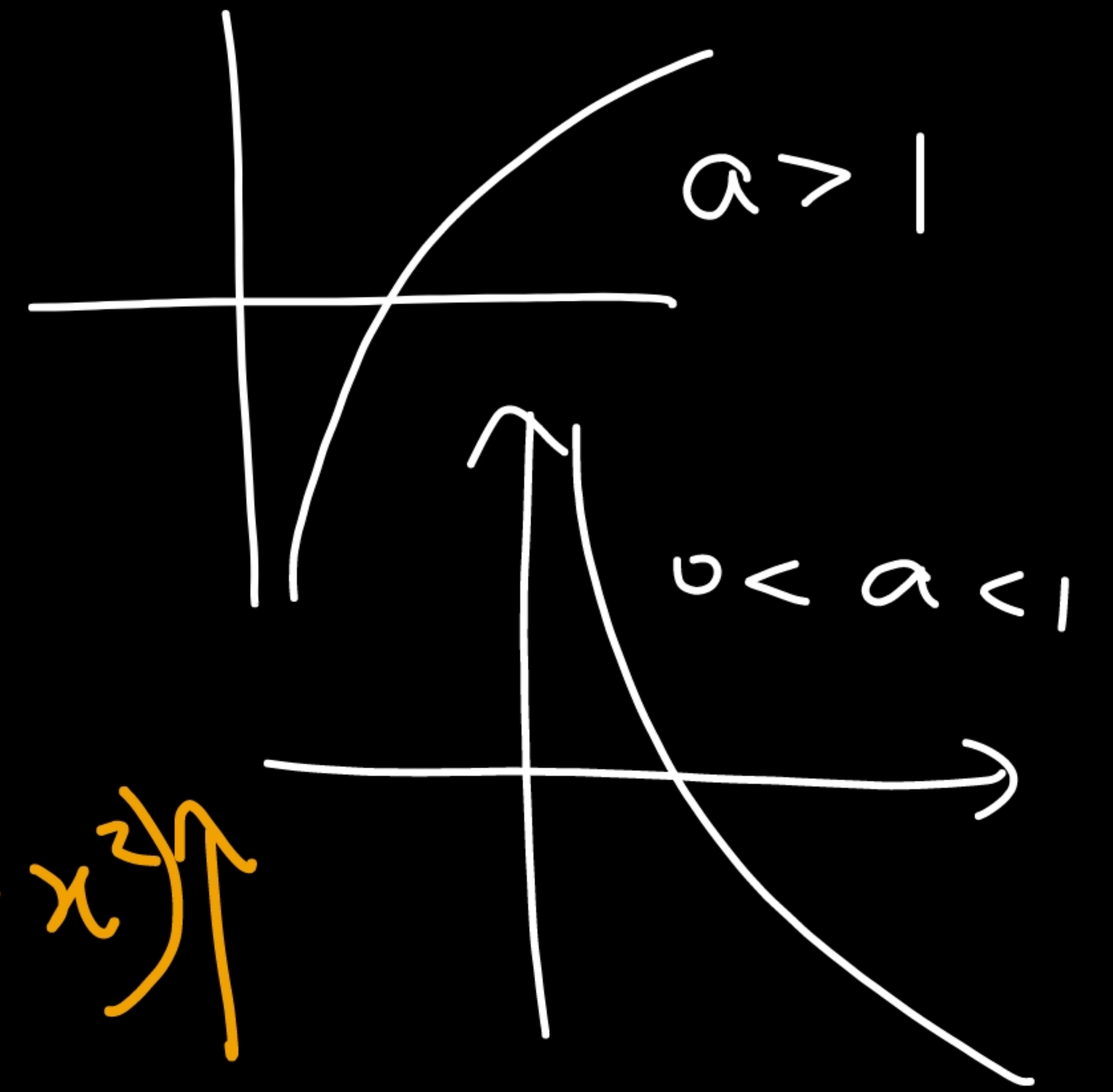
Ques

If the range of values of 'a' for which $f(x) = \log_a(4ax - x^2)$

is strictly increasing for $x \in \left[\frac{3}{2}, 2\right]$ is $[p, q] \cup (r, \infty)$

then determine the value of $(2p + 4q + r)$.

 5



Sol

$$f(x) = \log_a(4ax - x^2)$$

$$x \in \left[\frac{3}{2}, 2\right]$$

↑
inc.

$$a \in (p, q] \cup (r, \infty)$$

C-1
 $a > 1 \rightarrow f(x) \uparrow^{inc} \Rightarrow (4ax - x^2) \uparrow$

C-2
 $0 < a < 1 \rightarrow f(x) \uparrow^{inc} \Rightarrow (4ax - x^2) \downarrow$

$a \in \left(\frac{1}{2}, \frac{3}{4}\right] \cup (1, \infty)$

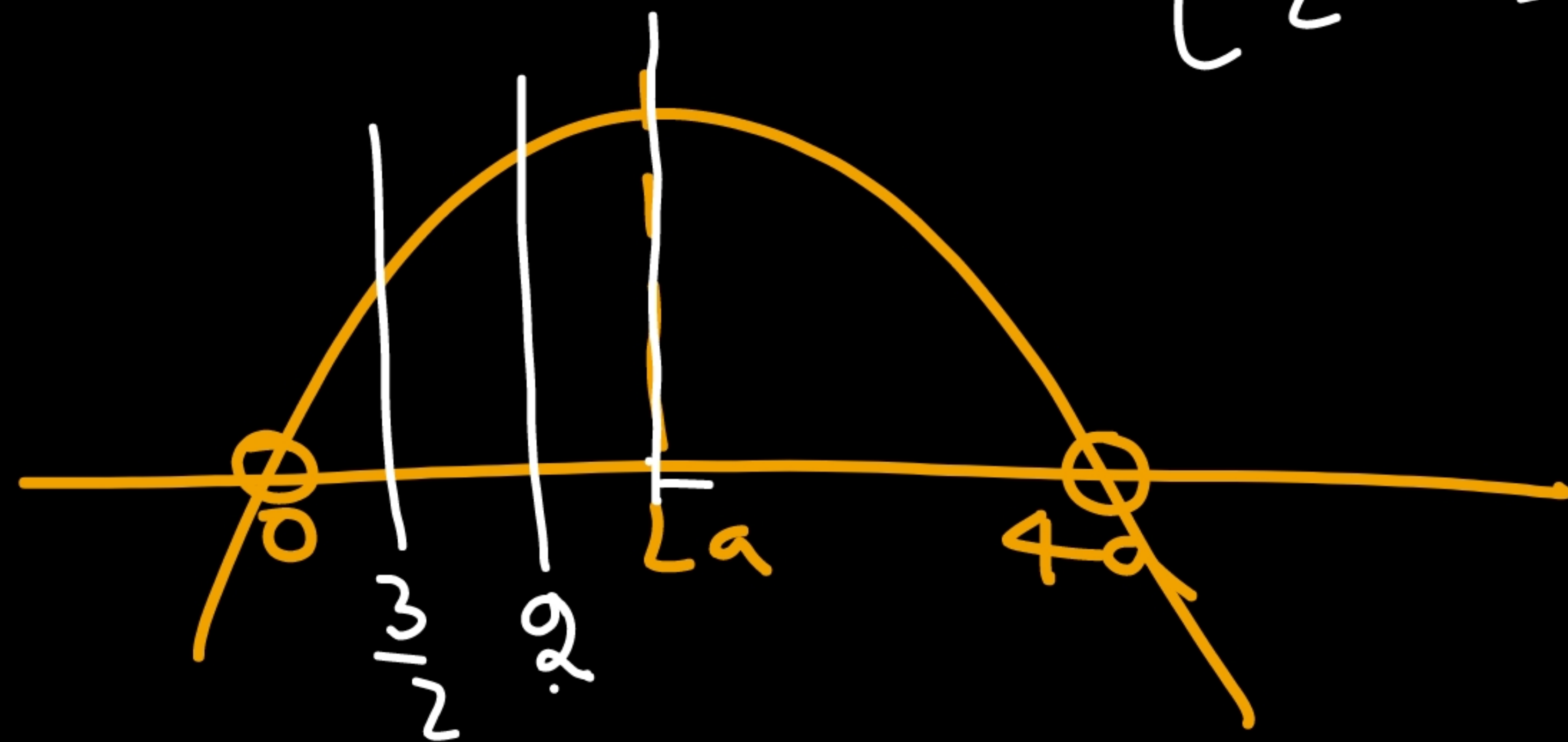
$p = \frac{1}{2}$	$\left\{ \begin{array}{l} 2p = 1 \\ 4q = 3 \\ r = 1 \end{array} \right.$
$q = \frac{3}{4}$	
$r = 1$	

5

C-1 $a > 1$

then $4ax - x^2$ must $\uparrow$

in $\left[\frac{3}{2}, 2\right]$



$2 \leq 2a$

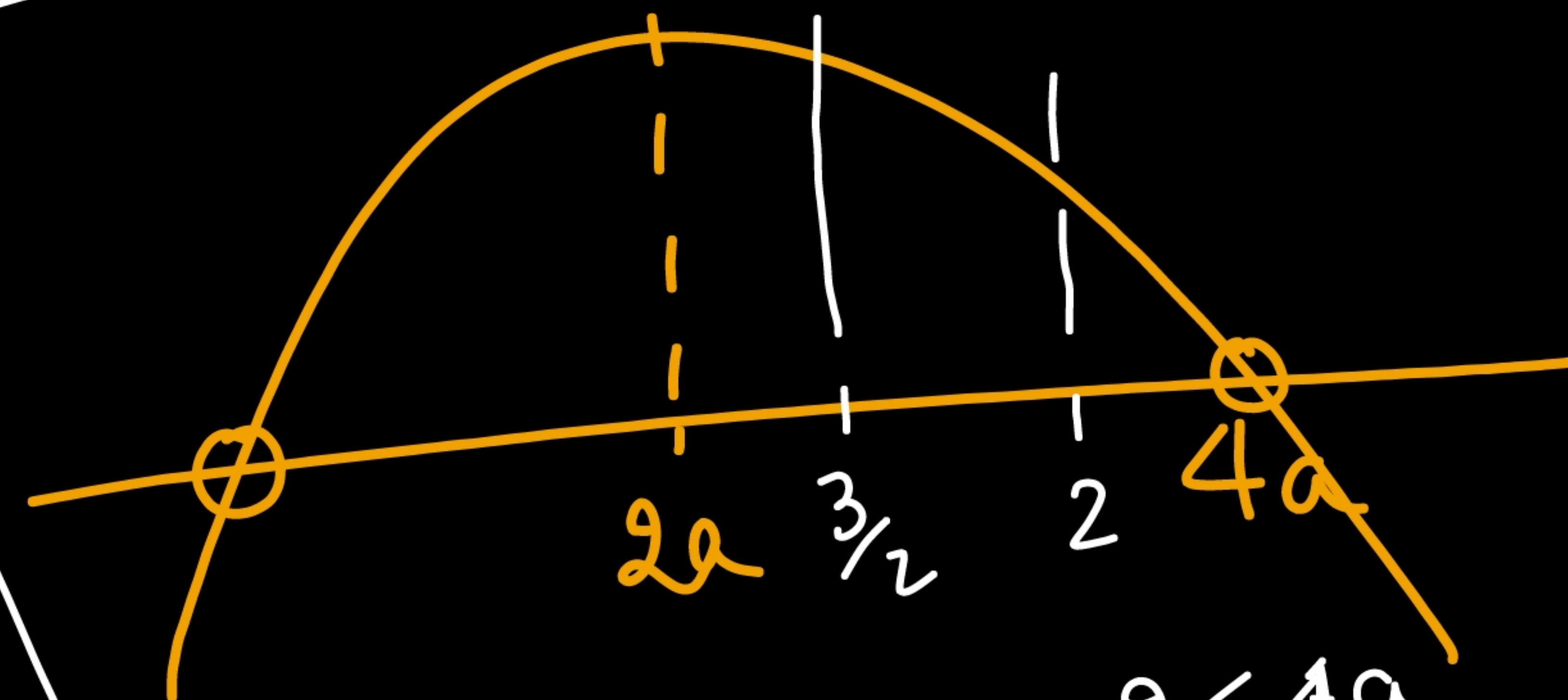
$a \geq 1$

$\cap \rightarrow a \in (1, \infty)$

C-2

$0 < a < 1$

$y = 4ax - x^2$ must $\downarrow$ in $\left[\frac{3}{2}, 2\right]$



$2 < 4a$

$2a \leq \frac{3}{2}$ $\otimes$

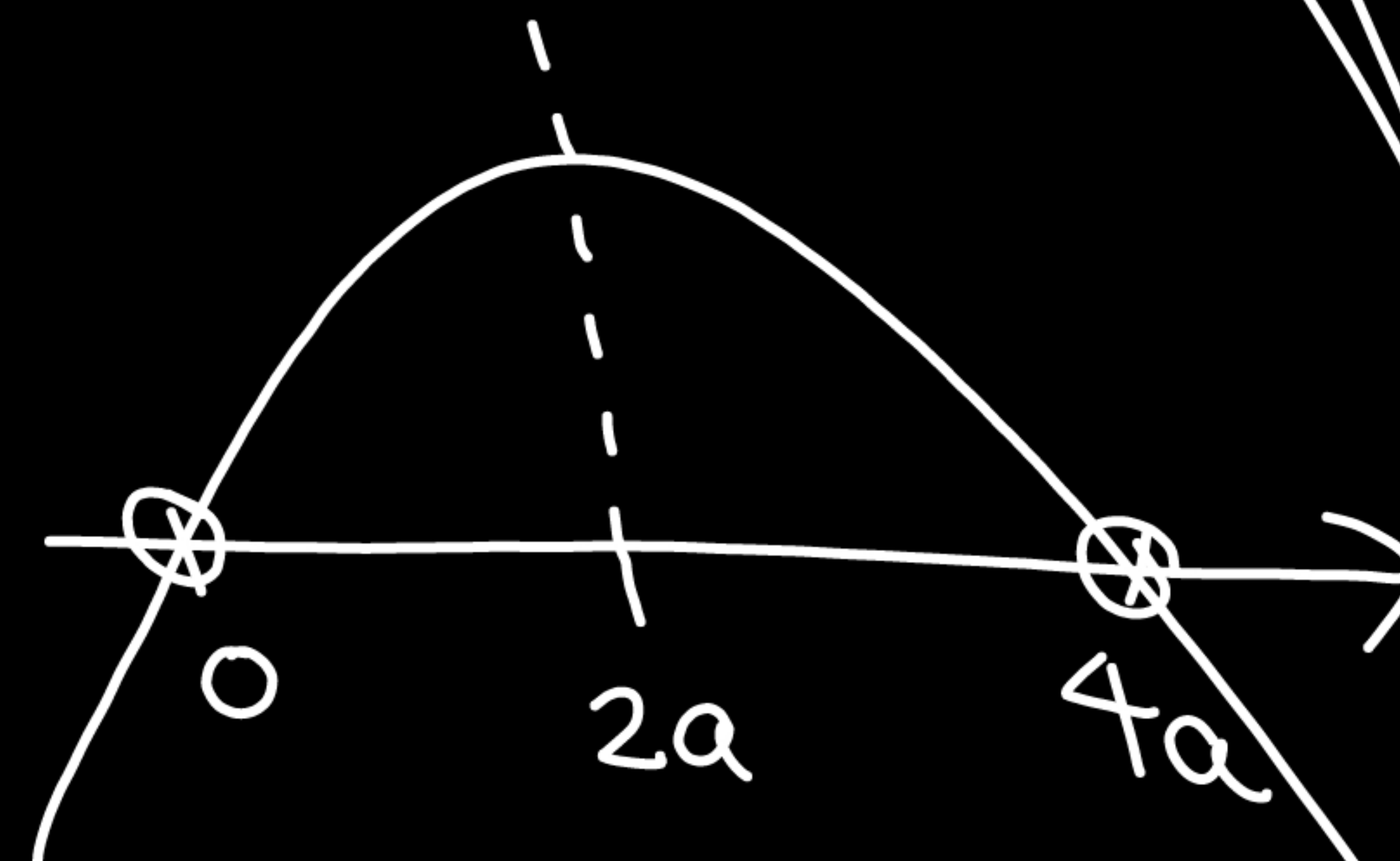
$a \leq \frac{3}{4}$ $\otimes$

$a > \frac{1}{2}$

$\cap \rightarrow a \in \left(\frac{1}{2}, \frac{3}{4}\right]$

$\cap \rightarrow a \in \left(\frac{1}{2}, \frac{3}{4}\right]$

$y = 4ax - x^2$
 $= x(4a - x)$



Que

Let N be the number of three distinct elements sets of positive integers such that their product is equal to 2310 then determine $\frac{N}{8}$.

5

$$\{\alpha, \beta, \gamma\}$$

$$N \rightarrow$$

$$\alpha, \beta, \gamma \in \mathbb{I}^+$$

α	β	γ
2310	...	...
2	5	77
6	5	77
5	6	77

$$\alpha \cdot \beta \cdot \gamma = 2310 = 2 \times 5 \times 3 \times 7 \times 11$$

$$\alpha \cdot \beta \cdot \gamma = 2 \times 3 \times 5 \times 7 \times 11$$

$$\frac{3^5 - 3}{3!} = \frac{3(81-1)}{6} = \frac{3 \times 80}{6} = 40$$

$$\frac{2}{8} = \frac{40}{8} = \boxed{5}$$

Que

Which of the following is/are true

(A) Let $g(x)$ be the inverse of $f(x)$ and $f(x)$ has domain $x \in [1, 5]$ where $f(1) = 2$ and $f(5) = 10$ then the

value of $\int_1^5 f(x) dx + \int_2^{10} g(x) dx$ is 48

~~(B)~~ Let $g(x)$ be the inverse of $f(x)$ and $f(x)$ has domain $x \in [1, 5]$ where $f(1) = 2$ and $f(5) = 10$ then the

~~value of $\int_1^5 f(x) dx + \int_2^{10} g(x) dx$ is 36~~

~~(C)~~ $\int_a^b x^{n-1} \left[\frac{(n-2)x^2 + (n-1)(a+b)x + nab}{(x+a)^2(x+b)^2} \right] dx = \frac{b^{n+1} - a^{n+1}}{2(a+b)}$

~~(D)~~ $\int_a^b x^{n-1} \left[\frac{(n-2)x^2 + (n-1)(a+b)x + nab}{(x+a)^2(x+b)^2} \right] dx = \frac{b^n - a^n}{2(a+b)}$

A

$$\int_a^b x^{n-1} \left(\frac{(n-2)x^2 + (n-1)(a+b)x + n \cdot ab}{(x+a)^2(x+b)^2} \right) dx$$

$$\left(\frac{f}{g} \right)' = \frac{f'g - f g'}{g^2}$$

$$= \int_a^b \left(\frac{n \cdot x^{n-1} \cdot (x+a)(x+b) - x^n \cdot (2x+a+b)}{((x+a)(x+b))^2} \right) dx$$

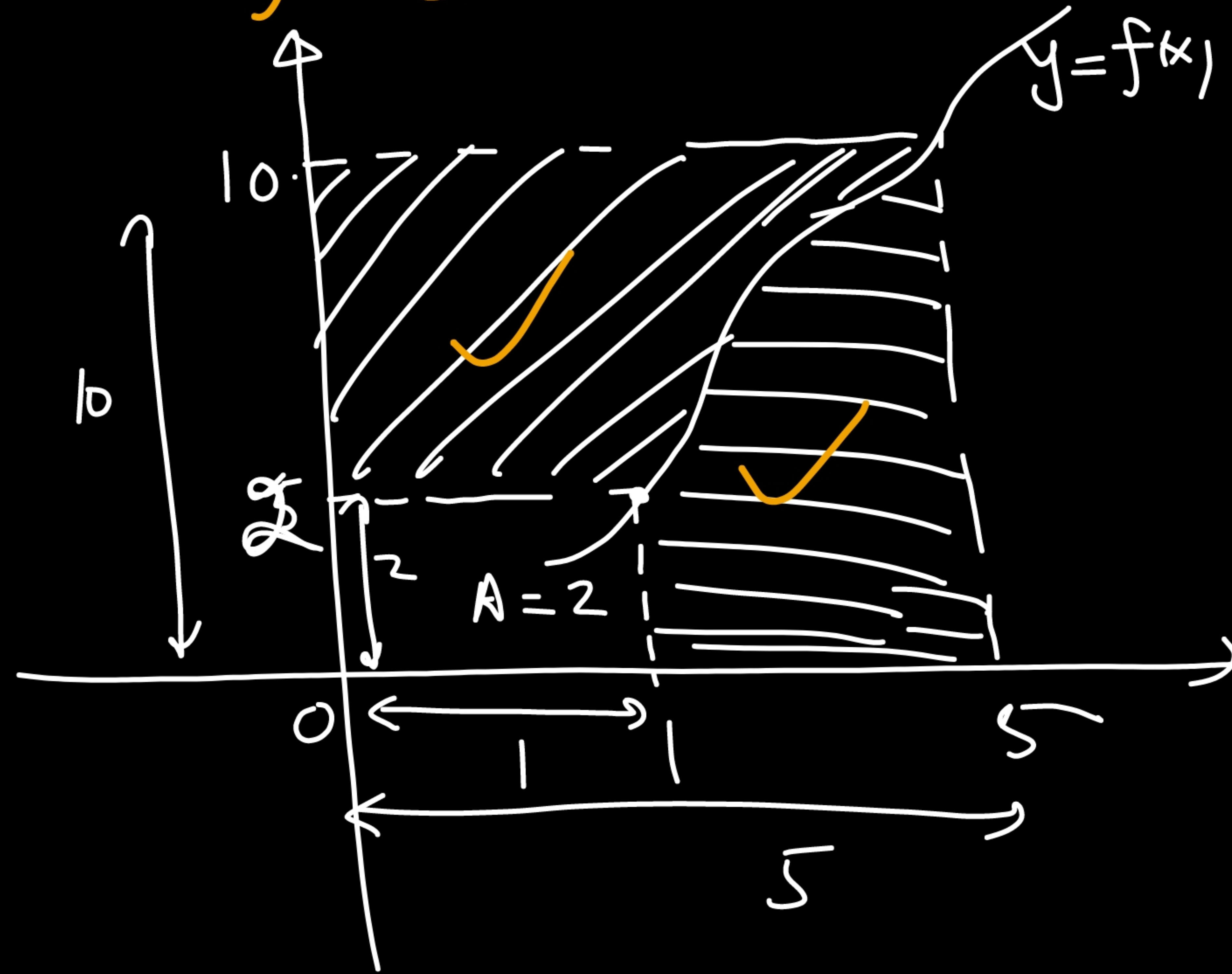
$$n(x^2 + (a+b)x + ab) - (2x^2 + (a+b)x)$$

$$n(x+a)(x+b) - x \cdot (2x + (a+b))$$

$$= \int_a^b \left(\frac{d}{dx} \left(\frac{x^n}{(x+a)(x+b)} \right) \right) dx = \left[\frac{x^n}{(x+a)(x+b)} \right]_a^b = \frac{b^n}{(b+a)2b} - \frac{a^n}{2a(a+b)}$$

$$= \frac{b^{n-1} - a^{n-1}}{2(a+b)} \quad \star$$

$$f: [1, 5] \rightarrow [2, 10]$$



$$50 - 2 = 48$$

$$\int_1^5 f(x) dx + \int_2^{10} g(x) dx = \int_2^{10} f^{-1}(x) dx$$